Patient demographics tables for Balder ML modeling

-

Overview

This document presents tables on patient demographics used for ML modeling. It primarily reports on GENIE data.

Loading data

Aggregation of data by cancer type

Data split by OncoTree codes

```
knitr::kable(outCntTbl, )
```

OT code	cancer type	cancer type detailed	number of patients	actionable events
LUAD	Lung Cancer	Lung Adenocarcinoma	23678	10286 (43.4%)
IDC	Breast Cancer	Breast Invasive Ductal Carcinoma	10117	4450 (44%)
BRCA	Breast Cancer	Invasive Breast Carcinoma	4050	1956 (48.3%)
MEL	Melanoma	Melanoma	4832	1556 (32.2%)
GIST	Gastrointestinal Stromal Tumor	Gastrointestinal Stromal Tumor	1896	1511 (79.7%)
COAD	Colorectal Cancer	Colon Adenocarcinoma	11806	1276 (10.8%)
THPA	Thyroid Cancer	Papillary Thyroid Cancer	1600	1070 (66.9%)
ILC	Breast Cancer	Breast Invasive Lobular Carcinoma	1635	1016 (62.1%)
PRAD	Prostate Cancer	Prostate Adenocarcinoma	7026	929 (13.2%)
AML	Acute myeloid leukemia	Acute myeloid leukemia	2825	868 (30.7%)
BLCA	Bladder Cancer	Bladder Urothelial Carcinoma	4498	777 (17.3%)
SKCM	Melanoma	Cutaneous Melanoma	2118	714 (33.7%)
NSCLC	Non-Small Cell Lung Cancer	Non-Small Cell Lung Cancer	2366	689 (29.1%)
ODG	Glioma	Oligodendroglioma	682	651 (95.5%)
HGSOC	Ovarian Cancer	High-Grade Serous Ovarian Cancer	4725	603 (12.8%)
ASTR	Glioma	Astrocytoma	821	559 (68.1%)
MDLC	Breast Cancer	Breast Mixed Ductal and Lobular Carcinoma	570	313 (54.9%)
BRCNOS	Breast Cancer	Breast Invasive Carcinoma, NOS	595	289 (48.6%)
BRCANOS	Breast Cancer	Breast Invasive Cancer, NOS	610	287 (47%)
COADREAD	Colorectal Cancer	Colorectal Adenocarcinoma	4145	283 (6.8%)
IHCH	Hepatobiliary Cancer	Intrahepatic Cholangiocarcinoma	1210	281 (23.2%)

Aggregation by sex and age

Data split by sex codes

```
knitr::kable(outSexTbl, )
```

cancer type detailed	male	female	uknown sex
Lung Adenocarcinoma	9184 (38.8%)	14303 (60.4%)	191 (0.8%)
Breast Invasive Ductal Carcinoma	95 (0.9%)	9926 (98.1%)	96 (0.9%)
Invasive Breast Carcinoma	44 (1.1%)	3966 (97.9%)	40 (1%)
Melanoma	2819 (58.3%)	1992 (41.2%)	21 (0.4%)
Gastrointestinal Stromal Tumor	1044 (55.1%)	837 (44.1%)	15 (0.8%)
Colon Adenocarcinoma	6072 (51.4%)	5622 (47.6%)	112 (0.9%)
Papillary Thyroid Cancer	666 (41.6%)	926 (57.9%)	8 (0.5%)
Breast Invasive Lobular Carcinoma	0 (0%)	1614 (98.7%)	21 (1.3%)
Prostate Adenocarcinoma	6918 (98.5%)	5 (0.1%)	103 (1.5%)
Acute myeloid leukemia	1607 (56.9%)	1187 (42%)	31 (1.1%)
Bladder Urothelial Carcinoma	3387 (75.3%)	1055 (23.5%)	56 (1.2%)
Cutaneous Melanoma	1343 (63.4%)	761 (35.9%)	14 (0.7%)
Non-Small Cell Lung Cancer	1139 (48.1%)	1211 (51.2%)	16 (0.7%)
Oligodendroglioma	378 (55.4%)	302 (44.3%)	2 (0.3%)
High-Grade Serous Ovarian Cancer	0 (0%)	4683 (99.1%)	42 (0.9%)

cancer type detailed	male	female	uknown sex
Astrocytoma	497 (60.5%)	318 (38.7%)	6 (0.7%)
Breast Mixed Ductal and Lobular Carcinoma	4 (0.7%)	565 (99.1%)	1 (0.2%)
Breast Invasive Carcinoma, NOS	3 (0.5%)	586 (98.5%)	6 (1%)
Breast Invasive Cancer, NOS	11 (1.8%)	596 (97.7%)	3 (0.5%)
Colorectal Adenocarcinoma	2293 (55.3%)	1845 (44.5%)	7 (0.2%)
Intrahepatic Cholangiocarcinoma	587 (48.5%)	613 (50.7%)	10 (0.8%)

Cancer type counts by study source

```
# #| echo: false
# #| include: false
# studyCnts <- perPatientOncoKBLevelSorted %>%
#   dplyr::group_by(ONCOTREE_CODE) %>%
#   dplyr::summarise(n_patients_total = n(),
#                    n_male=sum(SEX=="Male"),
#                    percMale = 100*n_male/n_patients_total,
#                    maleCntStr = paste0(n_male, " (", round(percMale, 1), "%)"),
#                    n_female=sum(SEX=="Female"),
#                    percFemale = 100*n_female/n_patients_total,
#                    femaleCntStr = paste0(n_female, " (", round(percFemale, 1), "%)"),
#                    n_unknown=sum(!(SEX=="Male" | SEX=="Female")),
#                    percUnknown = 100*n_unknown/n_patients_total,
#                    unknownCntStr = paste0(n_unknown, " (", round(percUnknown, 1), "%)"),
#                    mean_age=mean(AGE_AT_SEQ_REPORT))
#
#
# cTypeCntTbl <- cTypeThresh %>%
#   dplyr::left_join(cTypeRep[,c("ONCOTREE_CODE","CANCER_TYPE","CANCER_TYPE_DETAILED")],by=c("ONCOTREE_CODE")) %>%
#   dplyr::left_join(cTypeCnts,by=c("ONCOTREE_CODE")) %>%
#   dplyr::left_join(sexCnts,by=c("ONCOTREE_CODE")) %>%
#   dplyr::mutate(percActionable = 100*n_Level1_patients/n_patients,
#                 actionableEvents = paste0(n_Level1_patients, " (", round(percActionable, 1), "%)"))
#
# outSexTbl <- cTypeCntTbl[,c("CANCER_TYPE_DETAILED",
#                             "maleCntStr",
#                             "femaleCntStr",
#                             "unknownCntStr")]
# colnames(outSexTbl) <- c("cancer type detailed",
#                          "male",
#                          "female",
#                          "uknown sex")
```

Aggregation by Ancestry

```
raceByEthn <- data.frame(table(perPatientOncoKBLevelSorted$RACE,perPatientOncoKBLevelSorted$ETHNICITY))
```

Most common mutated genes for each cancer type

top five actionable genes for each cancer type

Top 20 most common actionable events by gene

```
knitr::kable(outGeneTbl[1:20,], )
```

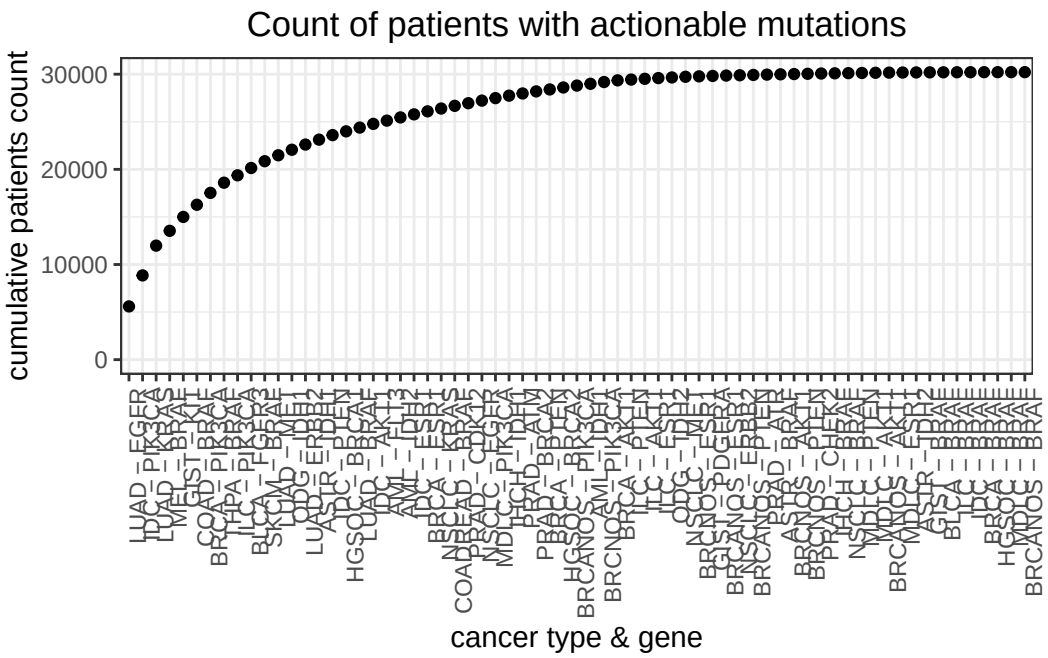
cancer type detailed	gene	count and perc of actionable events
Lung Adenocarcinoma	EGFR	5596 (54.4%)
Breast Invasive Ductal Carcinoma	PIK3CA	3259 (73.2%)
Lung Adenocarcinoma	KRAS	3129 (30.4%)
Melanoma	BRAF	1556 (100%)
Gastrointestinal Stromal Tumor	KIT	1462 (96.8%)
Colon Adenocarcinoma	BRAF	1276 (100%)
Invasive Breast Carcinoma	PIK3CA	1250 (63.9%)
Papillary Thyroid Cancer	BRAF	1070 (100%)
Breast Invasive Lobular Carcinoma	PIK3CA	775 (76.3%)
Bladder Urothelial Carcinoma	FGFR3	771 (99.2%)
Cutaneous Melanoma	BRAF	714 (100%)

cancer type detailed	gene	count and perc of actionable events
Lung Adenocarcinoma	MET	623 (6.1%)
Oligodendroglioma	IDH1	582 (89.4%)
Lung Adenocarcinoma	ERBB2	544 (5.3%)
Astrocytoma	IDH1	516 (92.3%)
Breast Invasive Ductal Carcinoma	PTEN	477 (10.7%)
High-Grade Serous Ovarian Cancer	BRCA1	397 (65.8%)
Lung Adenocarcinoma	BRAF	394 (3.8%)
Breast Invasive Ductal Carcinoma	AKT1	390 (8.8%)
Acute myeloid leukemia	FLT3	342 (39.4%)

Cumulative sum of mutations across the subset of cancer types

```
### calculate cumulative sum of mutations across
cTypeCntTbl$mutated_cumsum <- cumsum(cTypeCntTbl$n_gene_mutated)
cTypeCntTbl$perc_of_total <- 100* cTypeCntTbl$mutated_cumsum / sum(cTypeCntTbl$n_gene_mutated)
cTypeCntTbl$pair_str <- paste0(cTypeCntTbl$ONCOTREE_CODE, " - ", cTypeCntTbl$Hugo_Symbol)
cTypeCntTbl$pair_str <- factor(cTypeCntTbl$pair_str, levels=cTypeCntTbl$pair_str)
cMax <- max(cTypeCntTbl$mutated_cumsum)

outF <-  paste0(outDir,"/cumulative_mutation_counts_of_actionable_events.pdf")
#outF <-  paste0(outDir,"/cumulative_mutation_counts_of_actionable_events.png")
ggplot(cTypeCntTbl,aes(x=pair_str,y=mutated_cumsum))+
  geom_point() +
    labs(title = "Count of patients with actionable mutations",
       x = "cancer type & gene",
       y = "cumulative patients count") +
  ylim(0,cMax) +
  theme_bw()+
  theme(axis.text.x = element_text(angle = 90, hjust = 1))+
  #theme(axis.text.x = element_blank())+
  theme(plot.title = element_text(hjust = 0.5))
```




```
ggsave(outF,height = 6, width = 11)
```

```
print("portion of mutations acounted by ranked events")
```

```
[1] "portion of mutations acounted by ranked events"
```

```
print("top 10 events:")
```

```
[1] "top 10 events:"
```

```
print(cTypeCntTbl[10,c("perc_of_total")])
```

```
# A tibble: 1 x 1
  perc_of_total
          <dbl>
1          66.7
```

```
print("top 20 events:")
```

```
[1] "top 20 events:"
```

```
print(cTypeCntTbl[20,c("perc_of_total")])
```

```
# A tibble: 1 x 1
  perc_of_total
          <dbl>
1          83.1
```

```
print("top 30 events:")
```

```
[1] "top 30 events:"
```

```
print(cTypeCntTbl[30,c("perc_of_total")])
```

```
# A tibble: 1 x 1
  perc_of_total
          <dbl>
1          92.6
```

```
print("top 40 events:")
```

```
[1] "top 40 events:"
```

```
print(cTypeCntTbl[40,c("perc_of_total")])
```

```
# A tibble: 1 x 1
  perc_of_total
          <dbl>
1          97.9
```

Most common coding mutations for each cancer type

Ten most common mutations for each cancer type

```
knitr::kable(outMutTbl, )
```

cancer type detailed	gene	protein mutation	count and perc of actionable events
Lung Adenocarcinoma	KRAS	p.G12C	3123 (30.4%)
Lung Adenocarcinoma	EGFR	p.L858R	2145 (20.9%)
Lung Adenocarcinoma	EGFR	p.E746_A750del	1807 (17.6%)
Lung Adenocarcinoma	BRAF	p.V600E	394 (3.8%)
Lung Adenocarcinoma	ERBB2	p.Y772_A775dup	274 (2.7%)
Lung Adenocarcinoma	MET	p.X1010_splice	248 (2.4%)
Lung Adenocarcinoma	EGFR	p.G719A	197 (1.9%)
Lung Adenocarcinoma	EGFR	p.L861Q	189 (1.8%)
Lung Adenocarcinoma	EGFR	p.L747_P753delinsS	187 (1.8%)
Lung Adenocarcinoma	EGFR	p.L747_T751del	140 (1.4%)
Breast Invasive Ductal Carcinoma	PIK3CA	p.H1047R	1227 (27.6%)
Breast Invasive Ductal Carcinoma	PIK3CA	p.E545K	654 (14.7%)
Breast Invasive Ductal Carcinoma	AKT1	p.E17K	390 (8.8%)
Breast Invasive Ductal Carcinoma	PIK3CA	p.E542K	356 (8%)
Breast Invasive Ductal Carcinoma	PIK3CA	p.N345K	187 (4.2%)
Breast Invasive Ductal Carcinoma	PIK3CA	p.H1047L	172 (3.9%)
Breast Invasive Ductal Carcinoma	ESR1	p.D538G	119 (2.7%)

cancer type detailed	gene	protein mutation	count and perc of actionable events
Breast Invasive Ductal Carcinoma	ESR1	p.Y537S	90 (2%)
Breast Invasive Ductal Carcinoma	PIK3CA	p.C420R	70 (1.6%)
Breast Invasive Ductal Carcinoma	PTEN	p.T319*	30 (0.7%)
Invasive Breast Carcinoma	PIK3CA	p.H1047R	417 (21.3%)
Invasive Breast Carcinoma	PIK3CA	p.E545K	262 (13.4%)
Invasive Breast Carcinoma	AKT1	p.E17K	178 (9.1%)
Invasive Breast Carcinoma	PIK3CA	p.E542K	155 (7.9%)
Invasive Breast Carcinoma	ESR1	p.D538G	136 (7%)
Invasive Breast Carcinoma	PIK3CA	p.N345K	77 (3.9%)
Invasive Breast Carcinoma	ESR1	p.Y537S	72 (3.7%)
Invasive Breast Carcinoma	PIK3CA	p.H1047L	53 (2.7%)
Invasive Breast Carcinoma	PIK3CA	p.C420R	35 (1.8%)
Invasive Breast Carcinoma	ESR1	p.Y537N	27 (1.4%)
Melanoma	BRAF	p.V600E	1296 (83.3%)
Melanoma	BRAF	p.V600K	220 (14.1%)
Melanoma	BRAF	p.V600R	30 (1.9%)
Melanoma	BRAF	p.V600D	7 (0.4%)
Melanoma	BRAF	p.V600_K601delinsEI	1 (0.1%)
Melanoma	BRAF	p.V600_K601delinsEN	1 (0.1%)
Melanoma	BRAF	p.V600_S602delinsRKT	1 (0.1%)
Gastrointestinal Stromal Tumor	KIT	p.W557_K558del	143 (9.5%)
Gastrointestinal Stromal Tumor	KIT	p.A502_Y503dup	136 (9%)
Gastrointestinal Stromal Tumor	KIT	p.V559D	101 (6.7%)
Gastrointestinal Stromal Tumor	KIT	p.W557R	59 (3.9%)
Gastrointestinal Stromal Tumor	KIT	p.L576P	51 (3.4%)
Gastrointestinal Stromal Tumor	KIT	p.V560D	39 (2.6%)
Gastrointestinal Stromal Tumor	KIT	p.V559G	36 (2.4%)
Gastrointestinal Stromal Tumor	KIT	p.V654A	36 (2.4%)
Gastrointestinal Stromal Tumor	KIT	p.K642E	35 (2.3%)
Gastrointestinal Stromal Tumor	KIT	p.X550_splice	33 (2.2%)
Colon Adenocarcinoma	BRAF	p.V600E	1275 (99.9%)
Colon Adenocarcinoma	BRAF	p.T599_V600delinsRE	1 (0.1%)
Papillary Thyroid Cancer	BRAF	p.V600E	1070 (100%)
Breast Invasive Lobular Carcinoma	PIK3CA	p.H1047R	249 (24.5%)
Breast Invasive Lobular Carcinoma	PIK3CA	p.E545K	164 (16.1%)
Breast Invasive Lobular Carcinoma	PIK3CA	p.E542K	105 (10.3%)
Breast Invasive Lobular Carcinoma	AKT1	p.E17K	79 (7.8%)
Breast Invasive Lobular Carcinoma	PIK3CA	p.N345K	62 (6.1%)
Breast Invasive Lobular Carcinoma	PIK3CA	p.H1047L	43 (4.2%)
Breast Invasive Lobular Carcinoma	ESR1	p.D538G	28 (2.8%)
Breast Invasive Lobular Carcinoma	ESR1	p.Y537S	22 (2.2%)
Breast Invasive Lobular Carcinoma	PIK3CA	p.C420R	22 (2.2%)
Breast Invasive Lobular Carcinoma	ESR1	p.E380Q	14 (1.4%)
Prostate Adenocarcinoma	BRCA2	p.W1692Mfs*3	9 (1%)
Prostate Adenocarcinoma	ATR	p.X1913_splice	6 (0.6%)
Prostate Adenocarcinoma	FANCL	p.T367Ifs*13	6 (0.6%)
Prostate Adenocarcinoma	ATM	p.S2812Vfs*3	5 (0.5%)
Prostate Adenocarcinoma	CDK12	p.L123Tfs*4	5 (0.5%)
Prostate Adenocarcinoma	ATM	p.R337H	4 (0.4%)
Prostate Adenocarcinoma	ATM	p.X2929_splice	4 (0.4%)
Prostate Adenocarcinoma	BRCA2	p.L1227Qfs*5	4 (0.4%)
Prostate Adenocarcinoma	BRCA2	p.N1626Sfs*12	4 (0.4%)
Prostate Adenocarcinoma	BRCA2	p.T3033Nfs*11	4 (0.4%)
Acute myeloid leukemia	IDH2	p.R140Q	228 (26.3%)
Acute myeloid leukemia	IDH1	p.R132C	118 (13.6%)
Acute myeloid leukemia	IDH2	p.R172K	88 (10.1%)
Acute myeloid leukemia	IDH1	p.R132H	47 (5.4%)
Acute myeloid leukemia	FLT3	p.D835Y	42 (4.8%)
Acute myeloid leukemia	FLT3	p.D835H	30 (3.5%)
Acute myeloid leukemia	FLT3	p.D593_F594insNS	20 (2.3%)
Acute myeloid leukemia	IDH2	p.R140W	16 (1.8%)
Acute myeloid leukemia	IDH1	p.R132G	12 (1.4%)
Acute myeloid leukemia	FLT3	p.D835E	10 (1.2%)
Bladder Urothelial Carcinoma	FGFR3	p.S249C	497 (64%)
Bladder Urothelial Carcinoma	FGFR3	p.Y373C	143 (18.4%)
Bladder Urothelial Carcinoma	FGFR3	p.R248C	100 (12.9%)
Bladder Urothelial Carcinoma	FGFR3	p.G370C	30 (3.9%)
Bladder Urothelial Carcinoma	BRAF	p.V600E	6 (0.8%)
Bladder Urothelial Carcinoma	FGFR3	p.S371_Y373delinsGVC	1 (0.1%)
Cutaneous Melanoma	BRAF	p.V600E	591 (82.8%)
Cutaneous Melanoma	BRAF	p.V600K	93 (13%)
Cutaneous Melanoma	BRAF	p.V600R	20 (2.8%)
Cutaneous Melanoma	BRAF	p.V600D	5 (0.7%)
Cutaneous Melanoma	BRAF	p.V600G	4 (0.6%)
Cutaneous Melanoma	BRAF	p.V600_K601delinsEI	1 (0.1%)
Non-Small Cell Lung Cancer	KRAS	p.G12C	296 (43%)
Non-Small Cell Lung Cancer	EGFR	p.L858R	97 (14.1%)
Non-Small Cell Lung Cancer	EGFR	p.E746_A750del	90 (13.1%)
Non-Small Cell Lung Cancer	MET	p.X1010_splice	24 (3.5%)

cancer type detailed	gene	protein mutation	count and perc of actionable events
Non-Small Cell Lung Cancer	MET	p.X963_splice	24 (3.5%)
Non-Small Cell Lung Cancer	BRAF	p.V600E	19 (2.8%)
Non-Small Cell Lung Cancer	EGFR	p.L747_P753delinsS	11 (1.6%)
Non-Small Cell Lung Cancer	EGFR	p.L747_T751del	10 (1.5%)
Non-Small Cell Lung Cancer	EGFR	p.E746_S752delinsV	9 (1.3%)
Non-Small Cell Lung Cancer	ERBB2	p.Y772_A775dup	9 (1.3%)
Oligodendroglioma	IDH1	p.R132H	571 (87.7%)
Oligodendroglioma	IDH2	p.R172K	55 (8.4%)
Oligodendroglioma	IDH2	p.R172M	8 (1.2%)
Oligodendroglioma	IDH1	p.R132S	5 (0.8%)
Oligodendroglioma	IDH1	p.R132G	4 (0.6%)
Oligodendroglioma	IDH2	p.R172W	4 (0.6%)
Oligodendroglioma	IDH1	p.R132C	2 (0.3%)
Oligodendroglioma	IDH2	p.R172G	1 (0.2%)
Oligodendroglioma	IDH2	p.R172S	1 (0.2%)
High-Grade Serous Ovarian Cancer	BRCA1	p.E23Vfs*17	21 (3.5%)
High-Grade Serous Ovarian Cancer	BRCA1	p.Q1756Pfs*74	21 (3.5%)
High-Grade Serous Ovarian Cancer	BRCA1	p.R1699W	8 (1.3%)
High-Grade Serous Ovarian Cancer	BRCA1	p.X1559_splice	6 (1%)
High-Grade Serous Ovarian Cancer	BRCA1	p.S1253Rfs*10	5 (0.8%)
High-Grade Serous Ovarian Cancer	BRCA2	p.L2357Vfs*2	5 (0.8%)
High-Grade Serous Ovarian Cancer	BRCA1	p.C61G	4 (0.7%)
High-Grade Serous Ovarian Cancer	BRCA1	p.E1250*	4 (0.7%)
High-Grade Serous Ovarian Cancer	BRCA1	p.N1355Kfs*10	4 (0.7%)
High-Grade Serous Ovarian Cancer	BRCA1	p.X1731_splice	4 (0.7%)
Astrocytoma	IDH1	p.R132H	414 (74.1%)
Astrocytoma	IDH1	p.R132C	57 (10.2%)
Astrocytoma	BRAF	p.V600E	32 (5.7%)
Astrocytoma	IDH1	p.R132G	25 (4.5%)
Astrocytoma	IDH1	p.R132S	17 (3%)
Astrocytoma	IDH2	p.R172G	5 (0.9%)
Astrocytoma	IDH2	p.R172M	4 (0.7%)
Astrocytoma	IDH1	p.R132L	3 (0.5%)
Astrocytoma	IDH2	p.R172K	1 (0.2%)
Astrocytoma	IDH2	p.R172S	1 (0.2%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	p.H1047R	102 (32.6%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	p.E545K	51 (16.3%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	p.E542K	24 (7.7%)
Breast Mixed Ductal and Lobular Carcinoma	AKT1	p.E17K	18 (5.8%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	p.N345K	13 (4.2%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	p.H1047L	11 (3.5%)
Breast Mixed Ductal and Lobular Carcinoma	ESR1	p.D538G	8 (2.6%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	p.C420R	6 (1.9%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	p.E453K	5 (1.6%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	p.Q546K	5 (1.6%)
Breast Invasive Carcinoma, NOS	PIK3CA	p.H1047R	69 (23.9%)
Breast Invasive Carcinoma, NOS	PIK3CA	p.E545K	46 (15.9%)
Breast Invasive Carcinoma, NOS	AKT1	p.E17K	29 (10%)
Breast Invasive Carcinoma, NOS	PIK3CA	p.E542K	25 (8.7%)
Breast Invasive Carcinoma, NOS	ESR1	p.D538G	24 (8.3%)
Breast Invasive Carcinoma, NOS	ESR1	p.Y537S	9 (3.1%)
Breast Invasive Carcinoma, NOS	ESR1	p.E380Q	5 (1.7%)
Breast Invasive Carcinoma, NOS	PIK3CA	p.C420R	5 (1.7%)
Breast Invasive Carcinoma, NOS	PIK3CA	p.G1049R	5 (1.7%)
Breast Invasive Carcinoma, NOS	PIK3CA	p.N345K	5 (1.7%)
Breast Invasive Cancer, NOS	PIK3CA	p.H1047R	68 (23.7%)
Breast Invasive Cancer, NOS	PIK3CA	p.E545K	42 (14.6%)
Breast Invasive Cancer, NOS	PIK3CA	p.E542K	29 (10.1%)
Breast Invasive Cancer, NOS	ESR1	p.D538G	16 (5.6%)
Breast Invasive Cancer, NOS	AKT1	p.E17K	14 (4.9%)
Breast Invasive Cancer, NOS	PIK3CA	p.H1047L	12 (4.2%)
Breast Invasive Cancer, NOS	PIK3CA	p.N345K	12 (4.2%)
Breast Invasive Cancer, NOS	ESR1	p.Y537S	9 (3.1%)
Breast Invasive Cancer, NOS	ESR1	p.Y537N	5 (1.7%)
Breast Invasive Cancer, NOS	PIK3CA	p.E110del	4 (1.4%)
Colorectal Adenocarcinoma	BRAF	p.V600E	283 (100%)
Intrahepatic Cholangiocarcinoma	IDH1	p.R132C	186 (66.2%)
Intrahepatic Cholangiocarcinoma	IDH1	p.R132L	43 (15.3%)
Intrahepatic Cholangiocarcinoma	BRAF	p.V600E	23 (8.2%)
Intrahepatic Cholangiocarcinoma	IDH1	p.R132G	18 (6.4%)
Intrahepatic Cholangiocarcinoma	IDH1	p.R132S	9 (3.2%)
Intrahepatic Cholangiocarcinoma	IDH1	p.R132H	2 (0.7%)

Age distributions actionable vs non-actionable

```
print("subjects without an age value:")
```

```
[1] "subjects without an age value:"
```

```
print(table(is.na(perPatientOncoKBLevelSorted$AGE_AT_SEQ_REPORT)))
```

```
 FALSE   TRUE
201551  10636
```

```
iMatchCtype <- perPatientOncoKBLevelSorted$ONCOTREE_CODE %in% cTypeThresh$ONCOTREE_CODE
lvlEvents <- perPatientOncoKBLevelSorted[iMatchCtype & !is.na(perPatientOncoKBLevelSorted$AGE_AT_SEQ_REPORT),]
lvlEvents$actionabilityStr <- "no Level 1 event"
lvlEvents[lvlEvents$isOncoKbLevel1==1,"actionabilityStr"] <- "Level 1 event"
lvlEvents$actionabilityStr2 <- "Level1Event"
lvlEvents[lvlEvents$isOncoKbLevel1==1,"actionabilityStr2"] <- "NoLevel1Event"


lvlCenter <- lvlEvents %>%
  dplyr::group_by(ONCOTREE_CODE,actionabilityStr2) %>%
  dplyr::summarise(median_age=median(AGE_AT_SEQ_REPORT))
```

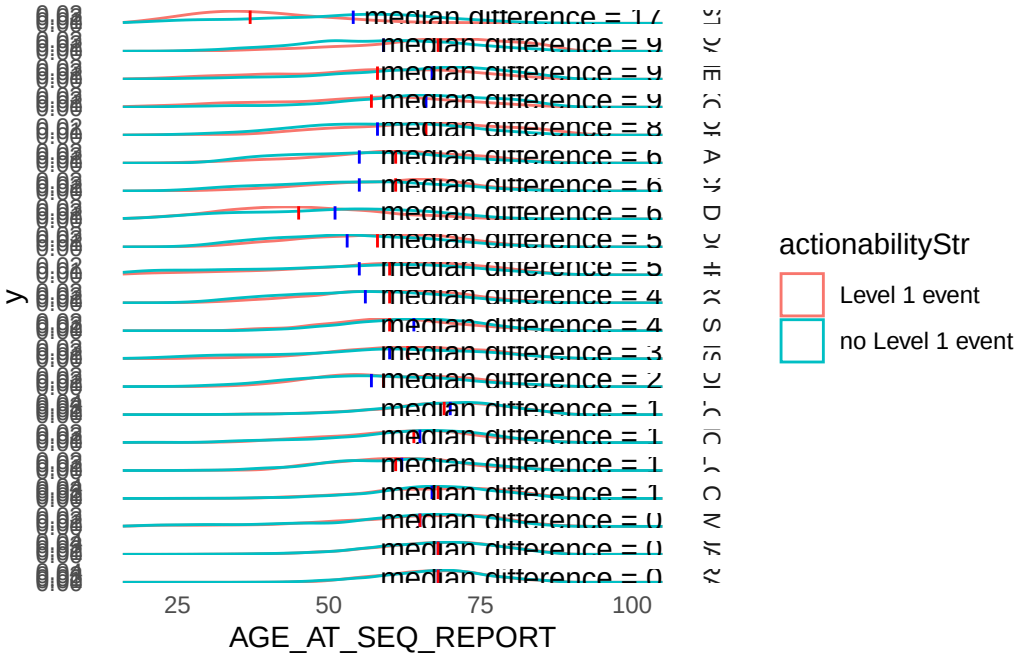
```
`summarise()` has grouped output by 'ONCOTREE_CODE'. You can override using the
`.groups` argument.
```

```
  #dplyr::summarise(mean_age=mean(AGE_AT_SEQ_REPORT))

lvlCenterLong <- lvlCenter %>%
  tidyr::pivot_wider(names_from = actionabilityStr2,values_from=median_age) %>%
  dplyr::mutate(ageDiff=abs(Level1Event-NoLevel1Event),
                diffString=paste0("median difference = ",round(ageDiff,1)),
                x=105,
                y=0.02) %>%
  dplyr::arrange(desc(ageDiff))

lvlEvents$ONCOTREE_CODE <- factor(lvlEvents$ONCOTREE_CODE,levels=lvlCenterLong$ONCOTREE_CODE)
lvlCenterLong$ONCOTREE_CODE <- factor(lvlCenterLong$ONCOTREE_CODE,levels=lvlCenterLong$ONCOTREE_CODE)

outF <-  paste0(outDir,"/age_difference_distribution_by_actionability.pdf")
ggplot(lvlEvents, aes(x = AGE_AT_SEQ_REPORT, color=actionabilityStr)) +
  geom_density(alpha=.4)+
  geom_vline(data = lvlCenterLong, aes(xintercept = Level1Event), color = "blue", linetype = "solid") +
  geom_vline(data = lvlCenterLong, aes(xintercept = NoLevel1Event), color = "red", linetype = "solid") +
  geom_text(data = lvlCenterLong, aes(x = x, y = y, label = diffString),
            inherit.aes = FALSE, hjust = 1, size = 4) +
  facet_grid(ONCOTREE_CODE~.,scale="free") +
    theme_minimal() +
  theme(
    strip.background = element_blank(),
    panel.grid = element_blank()
  )
```




```
ggsave(outF,height = 15, width = 9)
```

Sex differences among actionable vs non-actionable

```
print("subjects without an sex value:")
```

```
[1] "subjects without an sex value:"
```

```
print(table(is.na(perPatientOncoKBLevelSorted$SEX)))
```

```
 FALSE
212187
```

```
print(table(perPatientOncoKBLevelSorted$SEX))
```

```
 Female    Male   Other Unknown
 112211   97189       3    2784
```

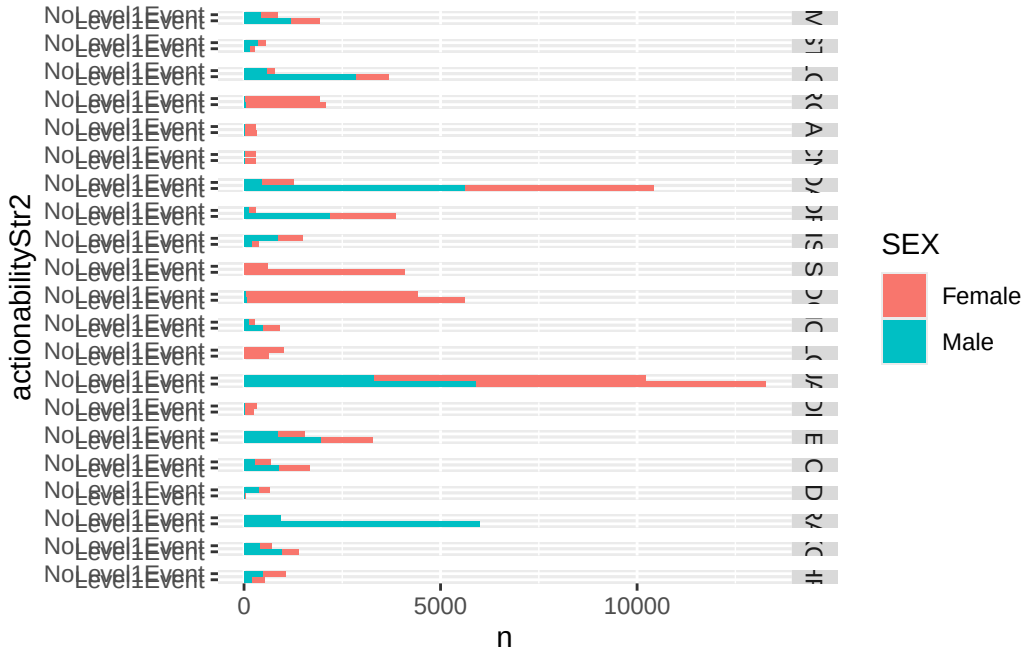
```
iMatchCtype <- perPatientOncoKBLevelSorted$ONCOTREE_CODE %in% cTypeThresh$ONCOTREE_CODE
iUnknown <- perPatientOncoKBLevelSorted$SEX == "Unknown" | perPatientOncoKBLevelSorted$SEX == "Other"
lvlEvents <- perPatientOncoKBLevelSorted[iMatchCtype & !iUnknown,]
lvlEvents$actionabilityStr <- "no Level 1 event"
lvlEvents[lvlEvents$isOncoKbLevel1==1,"actionabilityStr"] <- "Level 1 event"
lvlEvents$actionabilityStr2 <- "Level1Event"
lvlEvents[lvlEvents$isOncoKbLevel1==1,"actionabilityStr2"] <- "NoLevel1Event"


# lvlCenter <- lvlEvents %>%
#   dplyr::group_by(ONCOTREE_CODE,actionabilityStr2) %>%
#   dplyr::summarise(male=sum(SEX=="Male"),
#                    female=sum(SEX=="Female"))

lvlCenter <- lvlEvents %>%
  dplyr::group_by(ONCOTREE_CODE,actionabilityStr2,SEX) %>%
  dplyr::summarise(n=n())
```

```
`summarise()` has grouped output by 'ONCOTREE_CODE', 'actionabilityStr2'. You
can override using the `.groups` argument.
```

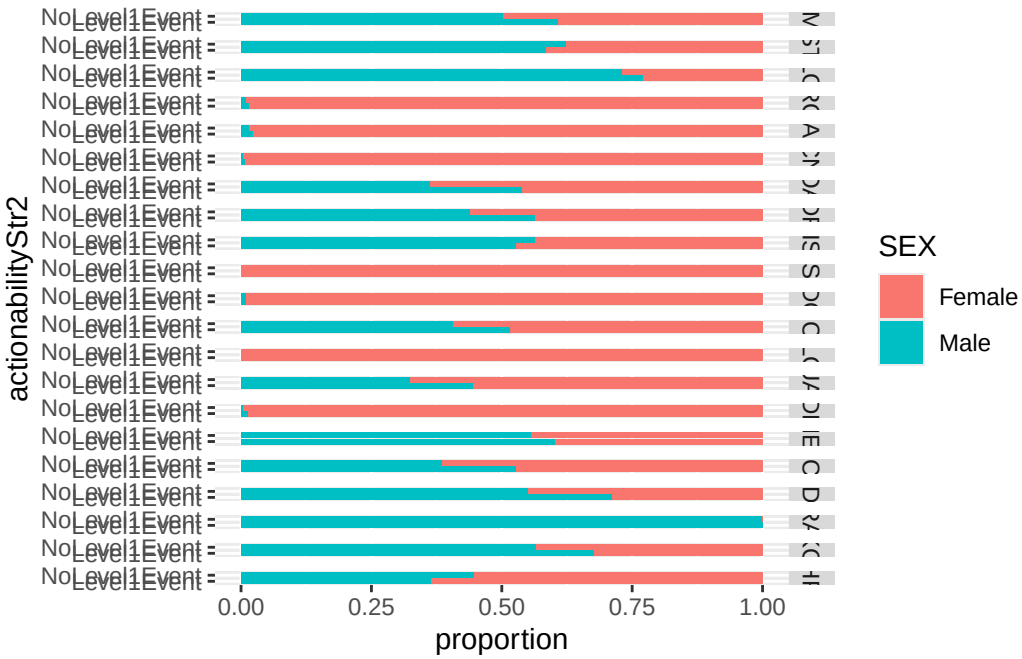
```
outF <-  paste0(outDir,"/sex_difference_distribution_by_actionability.pdf")
ggplot(lvlCenter,aes(x=actionabilityStr2,y=n,fill=SEX))+
  geom_bar(stat = "identity") +
  coord_flip() +
  facet_grid(ONCOTREE_CODE~.,scale="free")
```




```
ggsave(outF,height = 15, width = 9)

lvlProportion <- lvlCenter %>%
  dplyr::group_by(ONCOTREE_CODE,actionabilityStr2) %>%
  dplyr::mutate(proportion=n/sum(n))

outF <-  paste0(outDir,"/sex_difference_proportion_by_actionability.pdf")
ggplot(lvlProportion,aes(x=actionabilityStr2,y=proportion,fill=SEX))+
  geom_bar(stat = "identity") +
  coord_flip() +
  facet_grid(ONCOTREE_CODE~.,scale="free")
```




```
ggsave(outF,height = 15, width = 6)
```

Sample type differences among actionable vs non-actionable

```
print("subjects without an sample type value:")
```

```
[1] "subjects without an sample type value:"
```

```
print(table(is.na(perPatientOncoKBLevelSorted$SAMPLE_TYPE)))
```

```
 FALSE
212187
```

```
print(table(perPatientOncoKBLevelSorted$SAMPLE_TYPE))
```

```
      Local Recurrence             Metastasis Not Applicable or Heme
                    29                   66724                    7658
         Not Collected                Primary            Unspecified
                 15789                  106781                   15206
```

```
iMatchCtype <- perPatientOncoKBLevelSorted$ONCOTREE_CODE %in% cTypeThresh$ONCOTREE_CODE
lvlEvents <- perPatientOncoKBLevelSorted[iMatchCtype,]

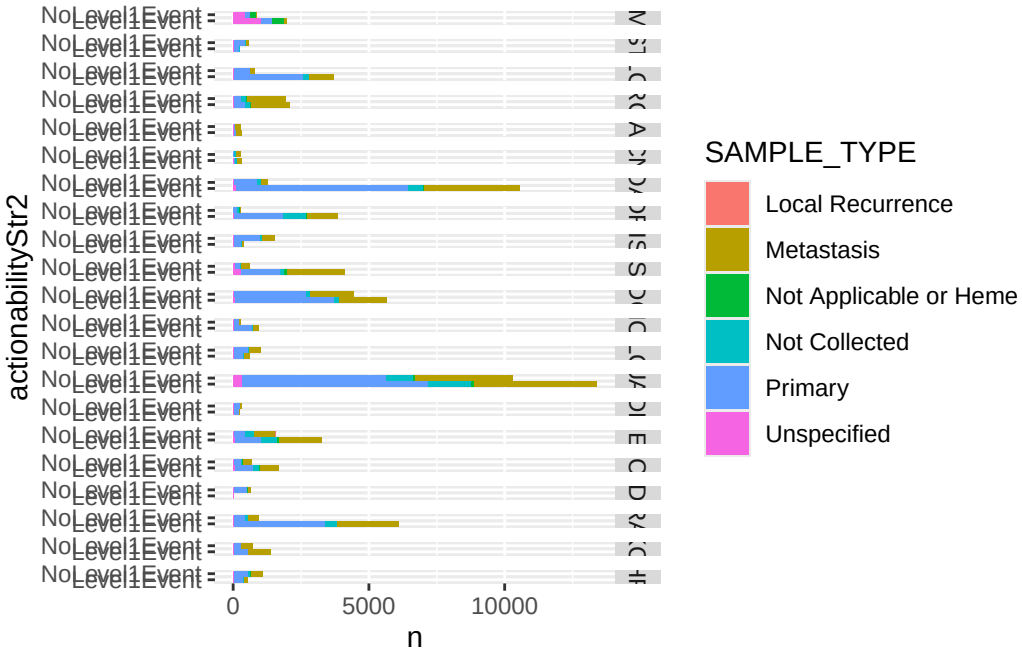

# iUnknown <- perPatientOncoKBLevelSorted$SEX == "Unknown" | perPatientOncoKBLevelSorted$SEX == "Other"
# lvlEvents <- perPatientOncoKBLevelSorted[iMatchCtype & !iUnknown,]

lvlEvents$actionabilityStr <- "no Level 1 event"
lvlEvents[lvlEvents$isOncoKbLevel1==1,"actionabilityStr"] <- "Level 1 event"
lvlEvents$actionabilityStr2 <- "Level1Event"
lvlEvents[lvlEvents$isOncoKbLevel1==1,"actionabilityStr2"] <- "NoLevel1Event"

lvlCenter <- lvlEvents %>%
  dplyr::group_by(ONCOTREE_CODE,actionabilityStr2,SAMPLE_TYPE) %>%
  dplyr::summarise(n=n())
```

```
`summarise()` has grouped output by 'ONCOTREE_CODE', 'actionabilityStr2'. You
can override using the `.groups` argument.
```

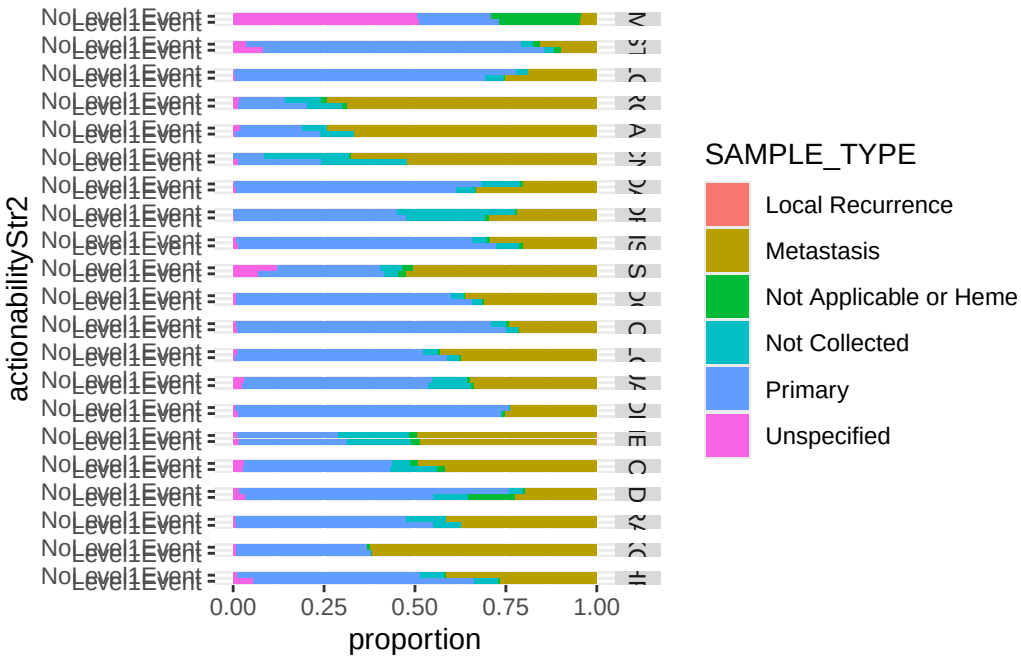
```
outF <-  paste0(outDir,"/sample_type_difference_distribution_by_actionability.pdf")
ggplot(lvlCenter,aes(x=actionabilityStr2,y=n,fill=SAMPLE_TYPE))+
  geom_bar(stat = "identity") +
  coord_flip() +
  facet_grid(ONCOTREE_CODE~.,scale="free")
```




```
ggsave(outF,height = 15, width = 9)

lvlProportion <- lvlCenter %>%
  dplyr::group_by(ONCOTREE_CODE,actionabilityStr2) %>%
  dplyr::mutate(proportion=n/sum(n))

outF <-  paste0(outDir,"/sample_type_difference_proportion_by_actionability.pdf")
ggplot(lvlProportion,aes(x=actionabilityStr2,y=proportion,fill=SAMPLE_TYPE))+
  geom_bar(stat = "identity") +
  coord_flip() +
  facet_grid(ONCOTREE_CODE~.,scale="free")
```




```
ggsave(outF,height = 15, width = 6)
```

Race differences among actionable vs non-actionable

```
print("subjects without an race value:")
```

```
[1] "subjects without an race value:"
```

```
print(table(is.na(perPatientOncoKBLevelSorted$RACE)))
```

```
 FALSE
212187
```

```
print(table(perPatientOncoKBLevelSorted$RACE))
```

```
                Asian                 Black  Native American                 Other
                11947                 11764                   395                 21459
Pacific Islander                 Unknown                 White
                  213                 30462                135947
```

```
iMatchCtype <- perPatientOncoKBLevelSorted$ONCOTREE_CODE %in% cTypeThresh$ONCOTREE_CODE
lvlEvents <- perPatientOncoKBLevelSorted[iMatchCtype,]

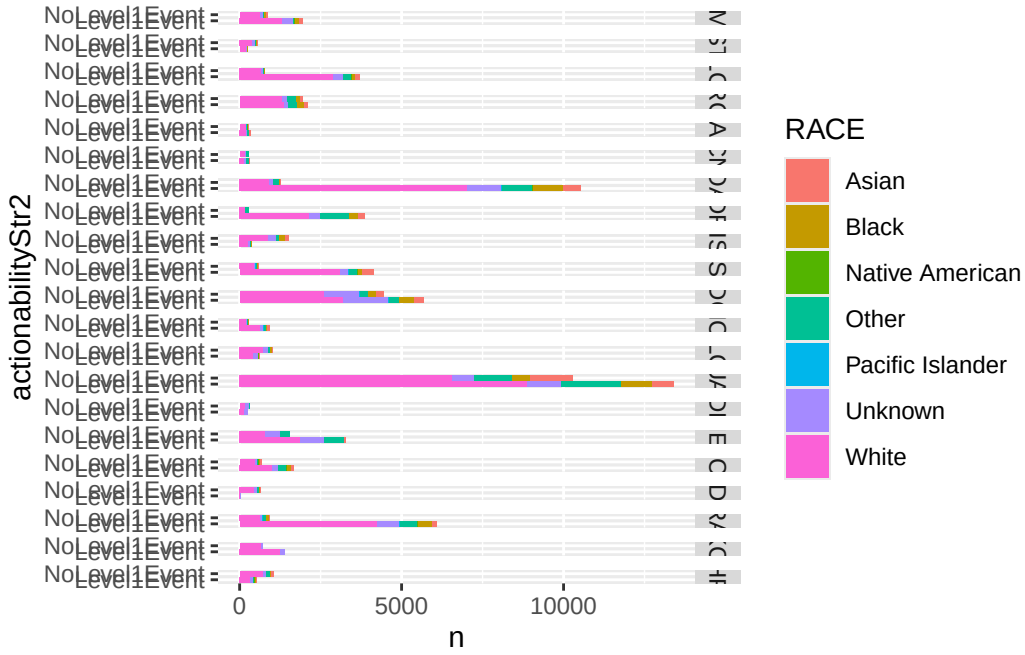

# iUnknown <- perPatientOncoKBLevelSorted$SEX == "Unknown" | perPatientOncoKBLevelSorted$SEX == "Other"
# lvlEvents <- perPatientOncoKBLevelSorted[iMatchCtype & !iUnknown,]

lvlEvents$actionabilityStr <- "no Level 1 event"
lvlEvents[lvlEvents$isOncoKbLevel1==1,"actionabilityStr"] <- "Level 1 event"
lvlEvents$actionabilityStr2 <- "Level1Event"
lvlEvents[lvlEvents$isOncoKbLevel1==1,"actionabilityStr2"] <- "NoLevel1Event"

lvlCenter <- lvlEvents %>%
  dplyr::group_by(ONCOTREE_CODE,actionabilityStr2,RACE) %>%
  dplyr::summarise(n=n())
```

```
`summarise()` has grouped output by 'ONCOTREE_CODE', 'actionabilityStr2'. You
can override using the `.groups` argument.
```

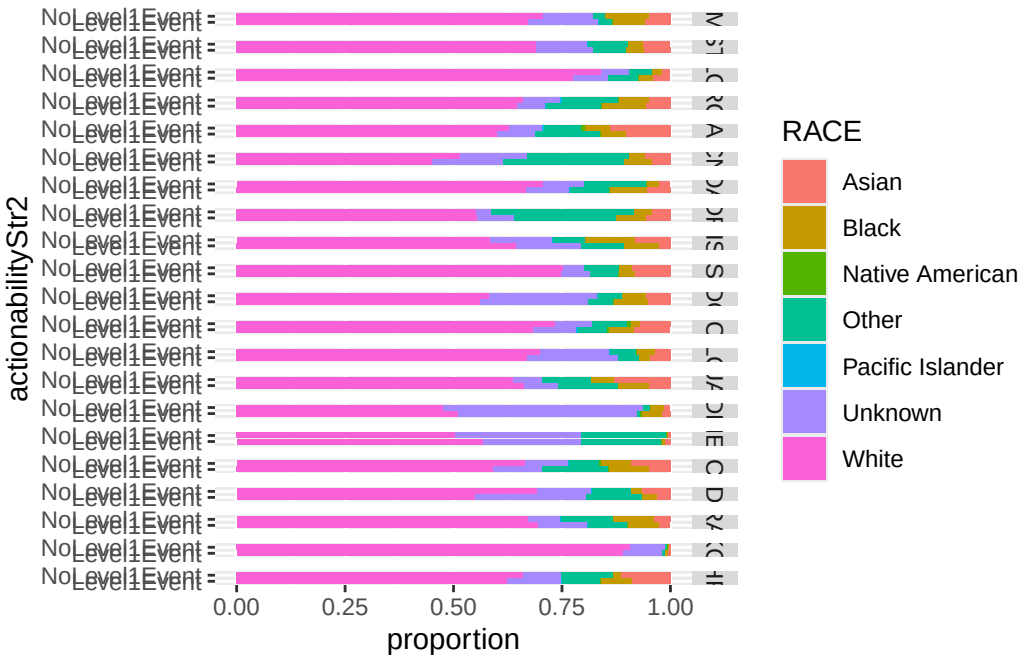
```
outF <-  paste0(outDir,"/race_difference_distribution_by_actionability.pdf")
ggplot(lvlCenter,aes(x=actionabilityStr2,y=n,fill=RACE))+
  geom_bar(stat = "identity") +
  coord_flip() +
  facet_grid(ONCOTREE_CODE~.,scale="free")
```




```
ggsave(outF,height = 15, width = 9)

lvlProportion <- lvlCenter %>%
  dplyr::group_by(ONCOTREE_CODE,actionabilityStr2) %>%
  dplyr::mutate(proportion=n/sum(n))

outF <-  paste0(outDir,"/race_difference_proportion_by_actionability.pdf")
ggplot(lvlProportion,aes(x=actionabilityStr2,y=proportion,fill=RACE))+
  geom_bar(stat = "identity") +
  coord_flip() +
  facet_grid(ONCOTREE_CODE~.,scale="free")
```




```
ggsave(outF,height = 15, width = 6)
```